

Name **KEY**

GT ID **Evans**

CHEM 2312 LS

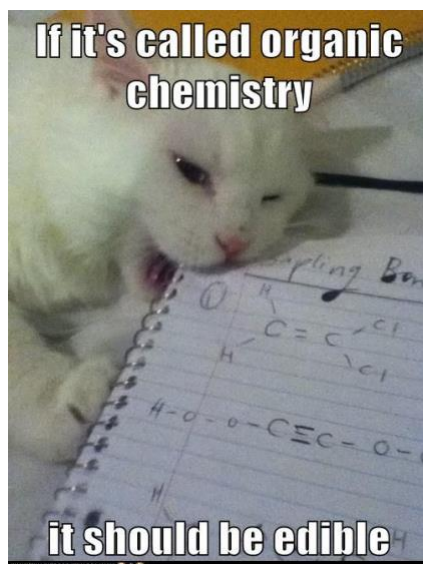
Organic Chemistry II

Midterm Examination 3

Take a deep breath. You got this.

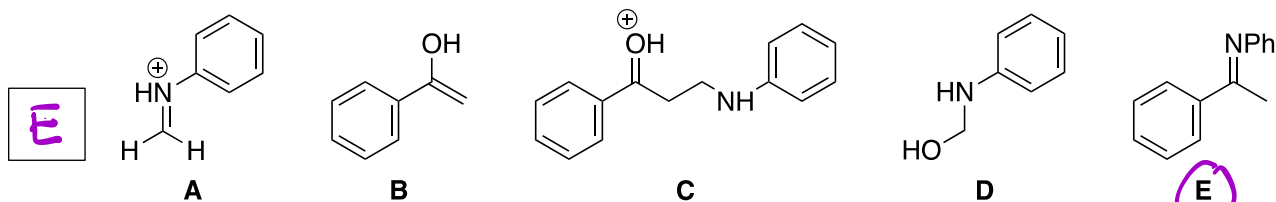
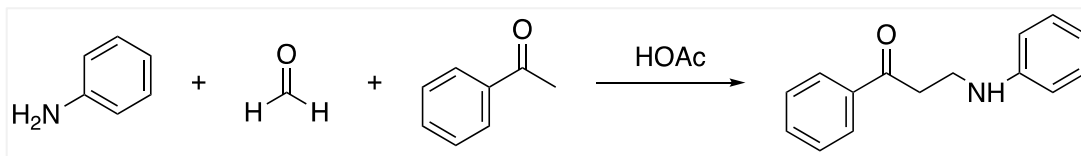
Do not open this exam until you are instructed to do so. Only a pencil or pen, scratch paper, and molecular model kit are permitted. You will have 50 minutes to complete the exam; it can be submitted at any time after 25 minutes have elapsed. Write all responses clearly in the boxes or spaces provided; work outside of designated areas will not be graded. On multiple-choice questions, write the letter of your choice in the box *and* circle your choice.

Ensure that configurations can be unambiguously assigned at all stereocenters (*R/S* or *E/Z*). Use “+ En” to indicate the presence of a racemic mixture when chiral products are generated from achiral starting materials.

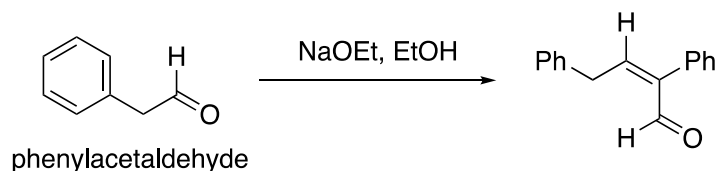


Multiple-choice Questions. Select the best response. Circle your choice and write its letter in the box. Each question is worth **4 points**.

1. Which of the following structures is *least* likely to be a reactive intermediate in the mechanism of the reaction below?



2. A student was interested in monitoring the progress of the dimerization of phenylacetaldehyde under basic conditions (see below) using spectroscopy. Which of the following changes is expected as the reaction proceeds?



- D

A. Disappearance of a signal in the infrared spectrum around 1700 cm^{-1}

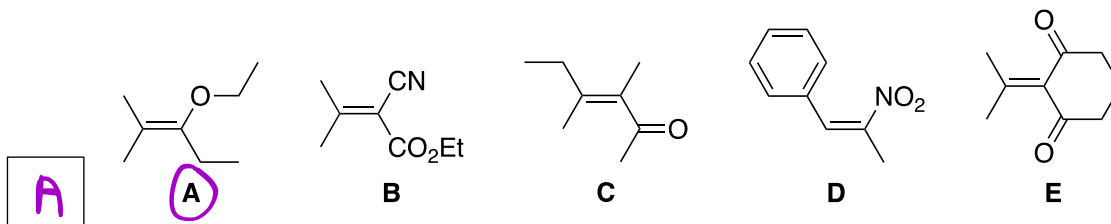
B. Disappearance of a signal in the infrared spectrum around 1500 cm^{-1}

C. Disappearance of a signal in the ^1H NMR spectrum around 10 ppm

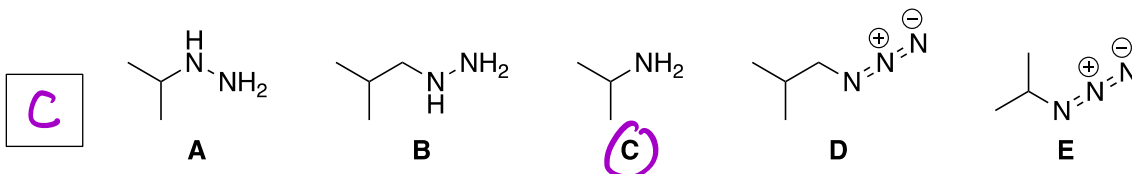
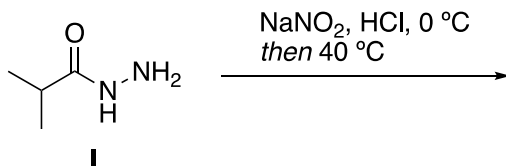
D. Appearance of a signal in the ^1H NMR spectrum around 6 ppm

E. Appearance of a signal in the ^1H NMR spectrum around 7.5 ppm

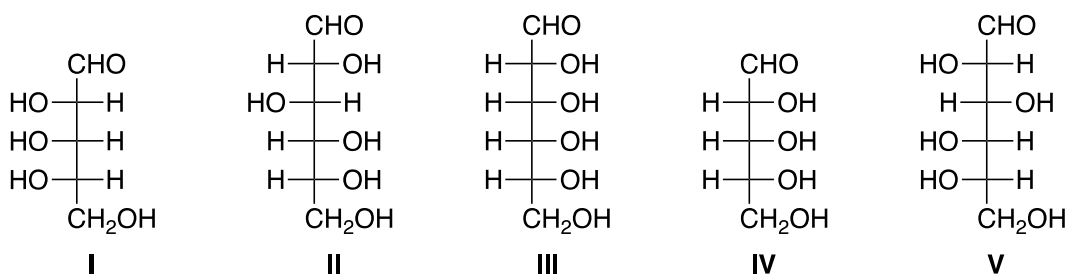
3. Which of the following alkenes *cannot* be prepared using an aldol-type condensation reaction involving a stabilized carbanion intermediate?



4. Acyl hydrazides such as **I** are converted to acyl azides in the presence of cold aqueous nitrous acid (HONO). Upon warming the reaction mixture to 40 °C, the acyl azide reacts further. What is the neutral organic product of the reaction below?



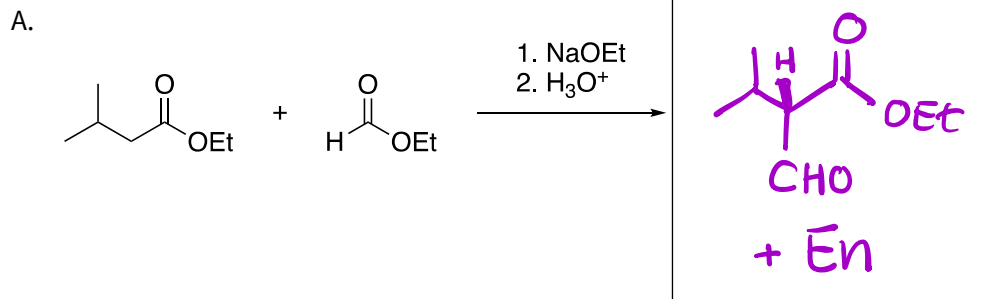
5. Consider the monosaccharides below. Of the listed pairs of compounds, which corresponds to a diastereomeric relationship?



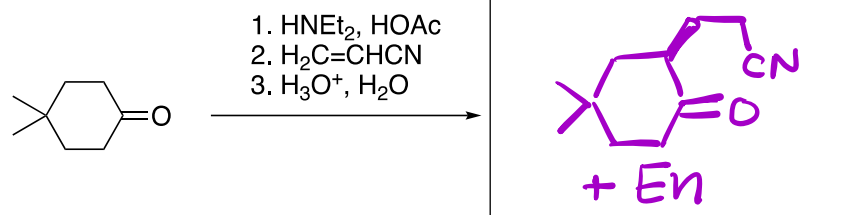
- B**
 A. **I** and **IV**
 B. **II** and **III**
 C. **II** and **IV**
 D. **II** and **V**
 E. **III** and **IV**

Other Questions. Place your response in the designated space for each question.

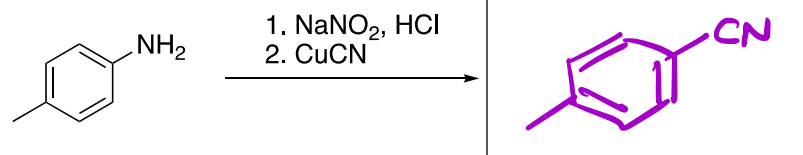
6. **20 pts.** Predict the neutral organic product of each reaction below.



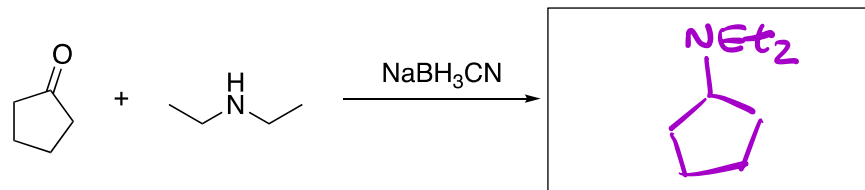
B.



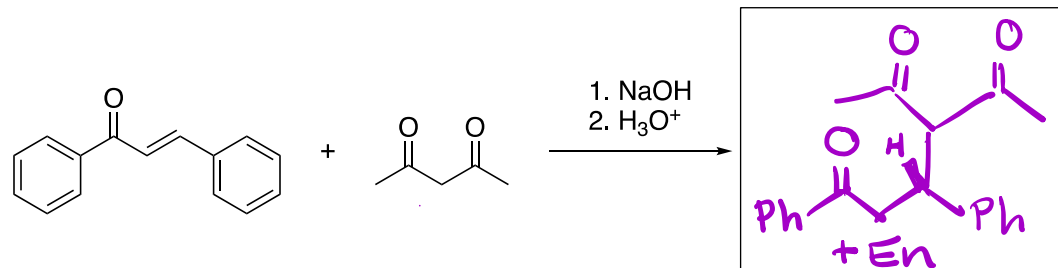
C.



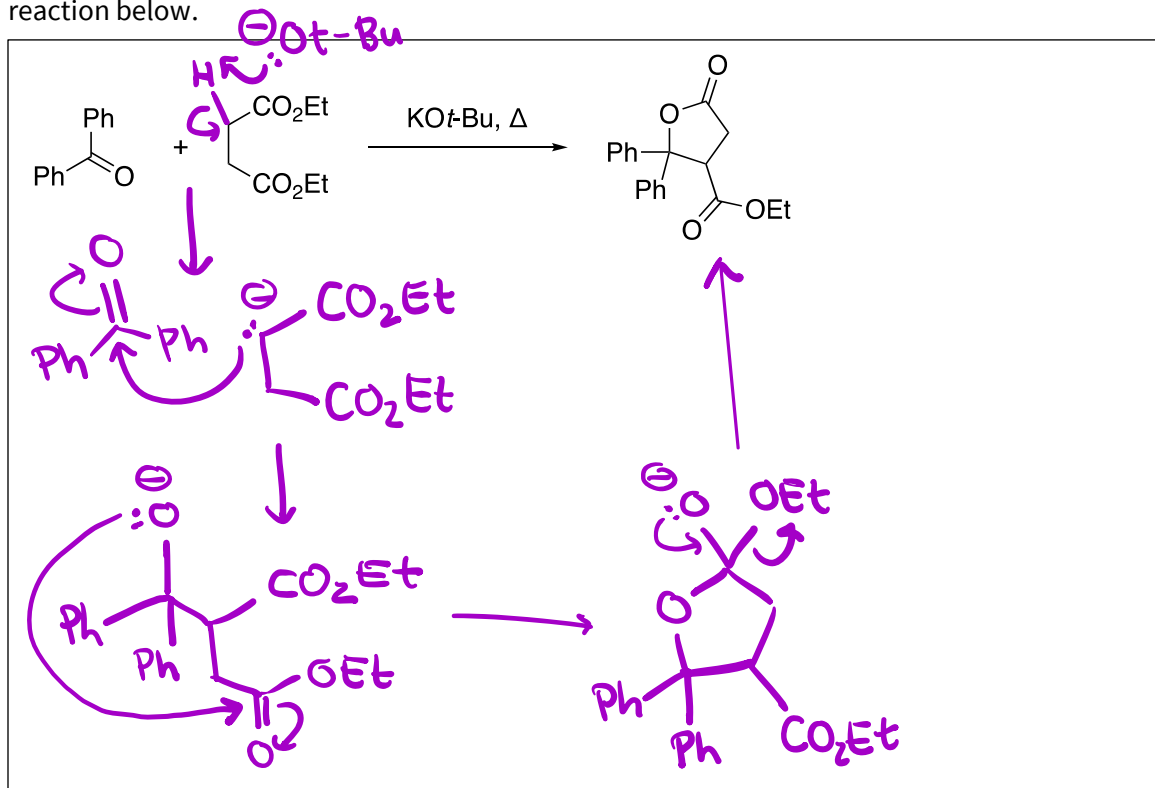
D.



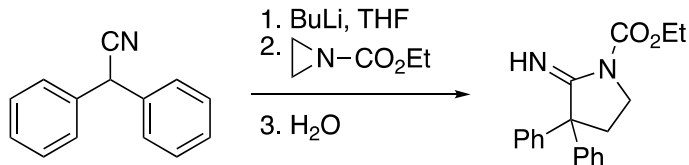
E.



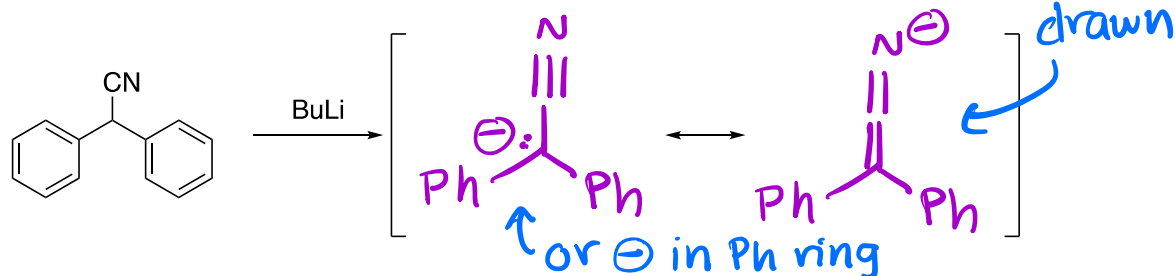
7. **12 pts.** Provide a mechanism with curved arrows and all reactive intermediates for the reaction below.



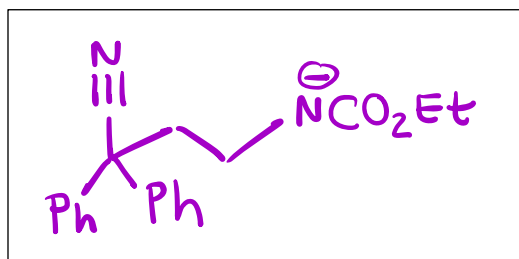
8. **12 pts.** Consider the mechanism of the reaction below.



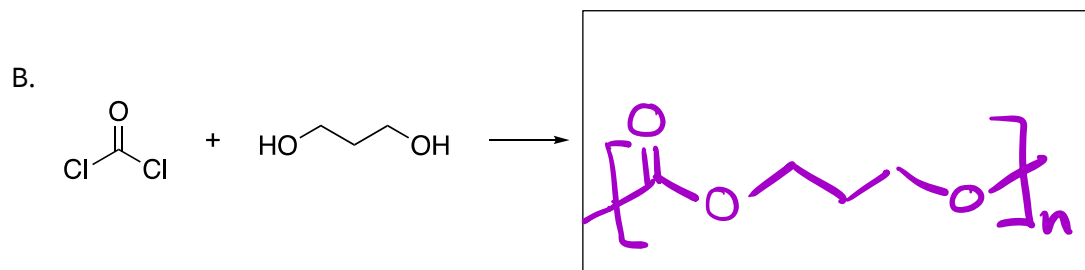
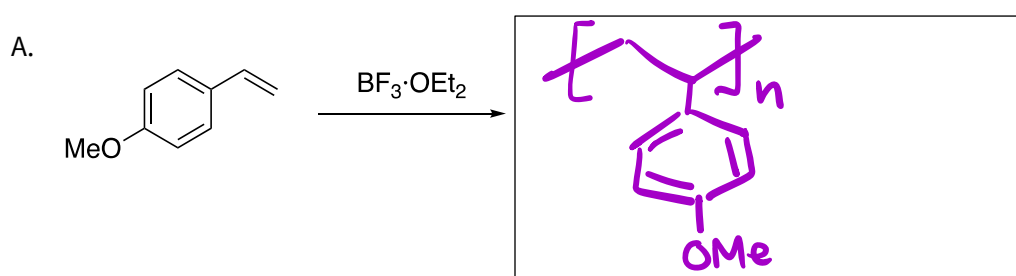
- A. *n*-Butyllithium (BuLi) is a strong base. Draw two resonance forms of the conjugate base of diphenylacetonitrile formed when BuLi deprotonates it. *One of your structures should be the most significant contributor to the resonance hybrid.*



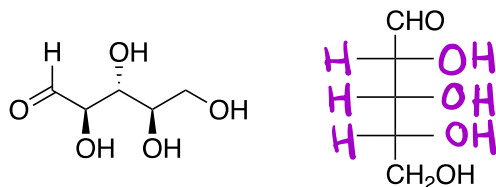
- B. The anion from part A engages with *N*-(ethoxycarbonyl)aziridine in an S_N2 step. Draw the structure of the anion formed in this step.



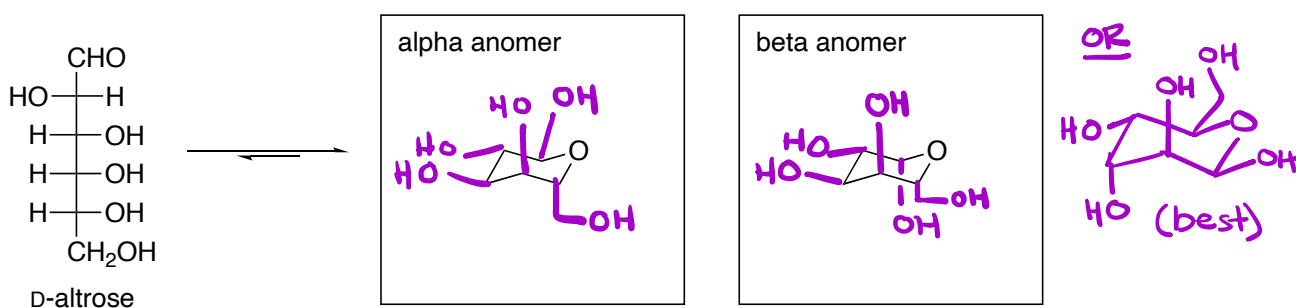
9. **12 pts.** Provide the structure of the neutral polymer formed in each of the reactions below. Include the structure of the repeating unit in brackets with a subscript n ; do not include end groups and don't worry about stereochemistry.



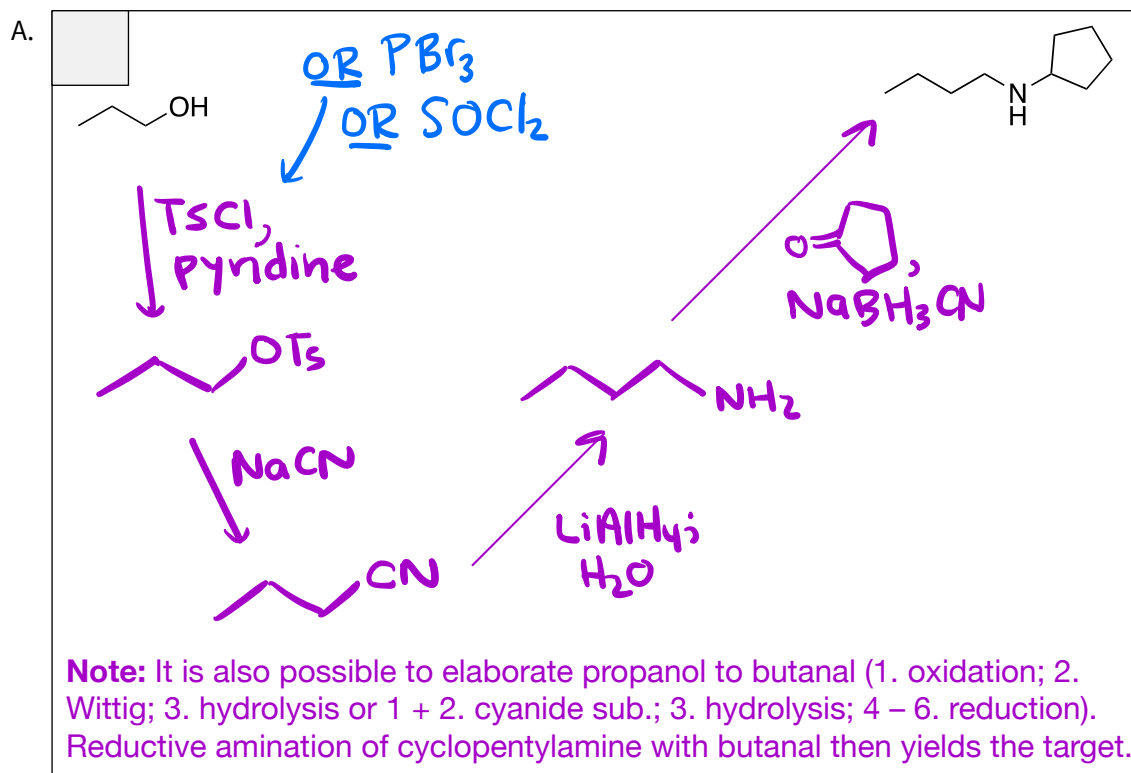
10. **6 pts.** Convert the structure below into a Fischer projection by adding H and OH groups to the "sticks" in the template below.



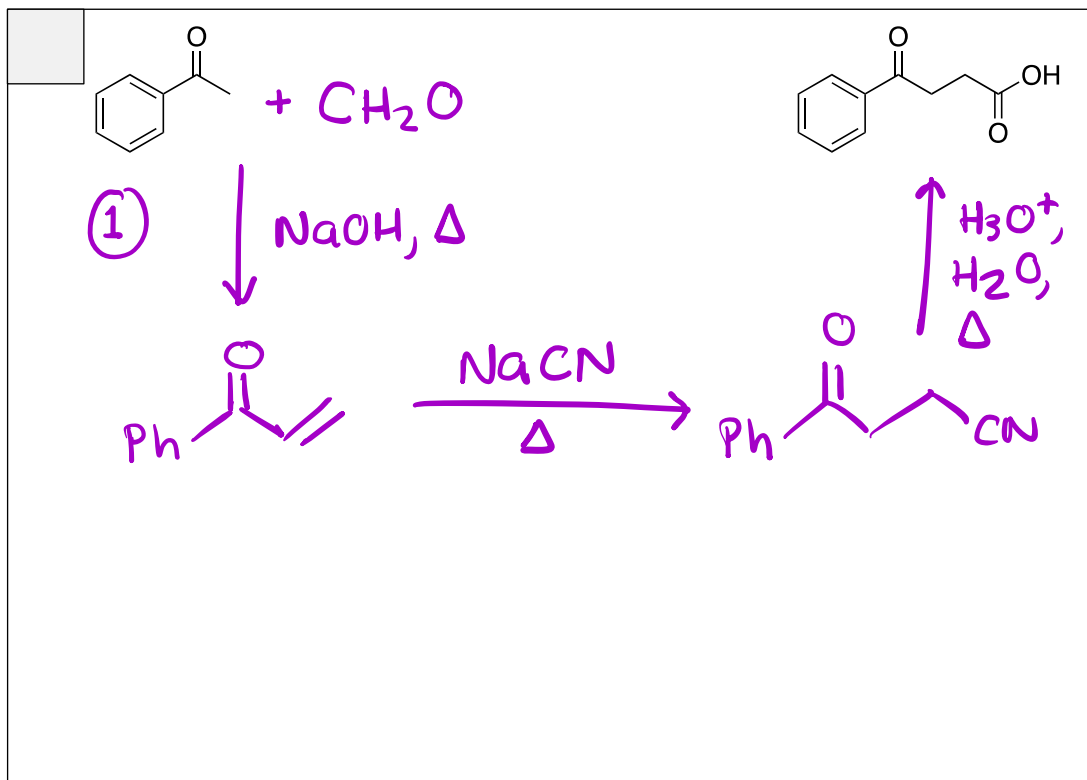
11. **8 pts.** In the boxes below, draw both anomers of the pyranose forms of D-altrose as their most stable chair conformers. Add -OH and -CH₂OH groups in the appropriate positions on the chair structures provided.



12. **10 pts.** Choose **one** of the two problems below, placing an X in the small gray box of your choice. Design a synthesis of the target (right) from the indicated starting material (left). You are welcome to use any existing reagents and sources of carbon to accomplish the synthesis, but it must start from the given starting material.



B.



There are several different ways to do this.

1. Aldol methenylation followed by warm cyanide addition (conjugate hydrocyanation) and then hydrolysis; shown above
2. Alpha-bromination and acetalization followed by malonic ester synthesis and deprotection. Protecting the ketone is needed (-1 if it isn't)
3. Enolate formation with LDA followed by epoxide opening and oxidation
4. Alkylation with a silyl-protected 2-haloalcohol followed by deprotection and oxidation, as in 3 (-1 for a potentially problematic alkylation)
5. Aldol condensation with protected hydroxyacetaldehyde followed by reduction, deprotection, and oxidation (-1 if lacking protection)

